

Dark Energy as Cascade Dynamics: No-Phantom Constraint, DESI Retrodiction, and Sealed Predictions for DESI DR3

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Abstract

The Discrete Stochastic Cascade model on finite graphs, developed in two prior works, is applied to the dark energy equation of state through a single physical identification: the cascade's global beta density maps to the dark energy density. Under this identification, five theorems follow directly from the cascade's proved guarantees, without any additional cosmological postulates; confronting them with data requires the phenomenological parameters $(w_0, \Omega_m, H_0 r_d)$, with $w_0 > -1$ enforced as a hard structural prior rather than fitted freely. Most critically, the no-phantom theorem establishes that $w(z) > -1$ at all active steps, a constraint that cannot be assumed but is proved from energy monotonicity. This constraint is tested against the DESI Year 1 Baryon Acoustic Oscillation dataset (April 2024). A blind retrodiction protocol (calibrating on three low-redshift points and predicting three withheld high-redshift points) yields a cascade chi-squared of 5.50 on unseen data, compared to 22.81 for CPL (which has one more free parameter and no phantom bound). The no-phantom constraint is the mechanism that prevents overfitting. A cross-release test training on DESI Year 1 and predicting DESI DR2 (March 2025) shows cascade performing at the Λ CDM level ($\chi^2 = 4.45$) while CPL achieves $\chi^2 = 2.17$ only via a phantom crossing at $z = 0.345$ that violates the Null Energy Condition. Six specific D_H/r_d predictions for DESI DR3 (expected 2027) are sealed in this document. If DESI DR3 confirms $w > -1$ at all redshifts and the CPL phantom preference weakens as error bars tighten, the no-phantom theorem and the cascade dark energy mapping survive a falsifiable empirical test; if DR3 finds $w < -1$ at high significance, the mapping is ruled out.

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1 Introduction

The dark energy equation of state remains the central open problem of observational cosmology. The DESI Year 1 results [3] report hints that $w(z)$ evolves with redshift rather than remaining fixed at -1 . The standard parameterization (CPL, [5, 6]) fits the data but predicts a phantom crossing ($w < -1$) that violates the Null Energy Condition and has no known physical mechanism.

This paper applies the Discrete Stochastic Cascade (DSC) model [1, 2] to the dark energy problem. The DSC model was not constructed for cosmology. It is a general framework for finite-resource stochastic dynamics on discrete graphs, with proved guarantees about energy depletion, structural hierarchy, and finite-time absorption. The cosmological application follows from a single identification (Section 3) and inherits five theorems already proved in the foundational works.

The key result is the no-phantom theorem (Theorem 4.2): the effective equation of state satisfies $w_{\text{eff}}(n) > -1$ at every active step. This is not assumed. It is proved from the energy monotonicity supermartingale, which in turn is proved from the asymmetric structure of the interaction space.

The present paper does not claim to outperform Λ CDM on current BAO data; on today's measurements the two are near-degenerate, and information criteria mildly favor Λ CDM (Section 6). Rather, it proposes a structurally constrained alternative whose distinguishing empirical consequence is the exclusion of phantom evolution, a commitment that DESI DR3 can falsify.

Priority statement. The sealed predictions in Section 7 were committed to a public repository at <https://github.com/Shiv2071/Discrete-cascade-model> on 21 June 2026, before DESI DR3 data exists. The git commit timestamp constitutes the independent priority record for these predictions.

2 The Cascade Model: Summary

The DSC model is defined on a finite graph $\mathcal{G} = (\mathcal{V}, \mathcal{A})$ with $P = |\mathcal{V}|$ sites. Each site p carries four coupled state variables:

$$\sigma(p, n) = (X(p, n), Y(p, n), \mathcal{E}(p, n), S(p, n))$$

where X, Y are two asymmetric excitation species, $\mathcal{E}$ is the local capacity energy (beta), and S is the structural state. The fundamental asymmetry is the *forbidden* Y - Y self-interaction channel ($\alpha_{YY} = 0$), which is proved in Part II [2] to be necessary for structural hierarchy.

The model evolves through three dynamical regimes determined by the ripple $F(p, n) = |S(p, n) - 2S(p, n - 1) + S(p, n - 2)|$:

Regime	Condition
Quiescent	$F(p, n) \leq C$
Leakage	$C < F(p, n) < C + \Delta$
Explosive	$F(p, n) \geq C + \Delta$

Three guarantees are proved in Part I [1]:

Theorem 2.1 (Energy Monotonicity, Part I Theorem 3.1). *The total capacity energy $\mathcal{E}_{\text{tot}}(n) = \sum_p \mathcal{E}(p, n)$ is a non-negative supermartingale:*

$$\mathbb{E}[\mathcal{E}_{\text{tot}}(n+1) \mid \mathcal{F}_n] \leq \mathcal{E}_{\text{tot}}(n)$$

and decreases strictly in expectation at every active step.

Theorem 2.2 (Finite Activity, Part I Theorem 3.2). *The total cumulative interaction count is almost surely finite, bounded by $\mathcal{E}_{\text{tot}}(0)/k_{\min}$.*

Theorem 2.3 (Almost Sure Absorption, Part I Theorem 3.3). *The system reaches an absorbing configuration in finite time with probability one, without fine-tuning to any critical point.*

3 Mapping to Dark Energy

Definition 3.1 (Cascade–Dark Energy Identification). Let $\rho_\beta(n)$ denote the mean beta density at cascade step n , and $\rho_{\text{DE}}(z)$ the dark energy density at cosmological redshift z . The single identification is:

$$\boxed{\rho_\beta(n) \longleftrightarrow \rho_{\text{DE}}(z(n))}$$

where $z(n)$ is the cascade-internal redshift defined by the Friedmann relation with the beta density as the dark energy source.

This identification is *interpretive*, not derived: it is a proposed correspondence between a variable of the cascade model and a cosmological quantity, analogous to identifying a lattice-model order parameter with a physical magnetisation. It is not obtained from cosmological field equations, and its justification is empirical: the constraints it induces must survive confrontation with data, which is the subject of Sections 5–7.

The identification itself introduces no new cascade-side parameters: the cascade’s internal dynamics generate the qualitative cosmological behavior (monotone depletion, $w > -1$, freeze-out) through this identification alone. Confronting the identification with data does require phenomenological parameters ($w_0, \Omega_m, H_0 r_d$), which are fitted in Sections 5–7; the structural content the cascade adds is the hard prior $w_0 > -1$ and the specific functional form of quiescent depletion.

4 Derived Theorems

Theorem 4.1 (Master Equation). *Under the cascade–dark energy identification, the effective equation of state at step n is:*

$$1 + w_{\text{eff}}(n) = \frac{D(n)}{3 H_{\text{an}}(n) \tau \rho_\beta(n)}$$

where $D(n) = \rho_\beta(n-1) - \rho_\beta(n) \geq 0$ is the depletion at step n , $H_{\text{an}}(n)$ is the analogous Hubble rate, and τ is the cascade time scale.

Theorem 4.2 (No-Phantom Theorem). *For all active steps n :*

$$w_{\text{eff}}(n) > -1$$

Proof. From Theorem 2.1, $D(n) = \rho_\beta(n-1) - \rho_\beta(n) > 0$ at every active step. Since $H_{\text{an}}(n) > 0$, $\tau > 0$, and $\rho_\beta(n) > 0$, the master equation gives $1 + w_{\text{eff}}(n) > 0$, hence $w_{\text{eff}}(n) > -1$. $\square$

Remark 4.3. This is the first derivation of a no-phantom constraint from discrete structural axioms. The constraint does not assume energy conditions; it proves them from the energy monotonicity supermartingale.

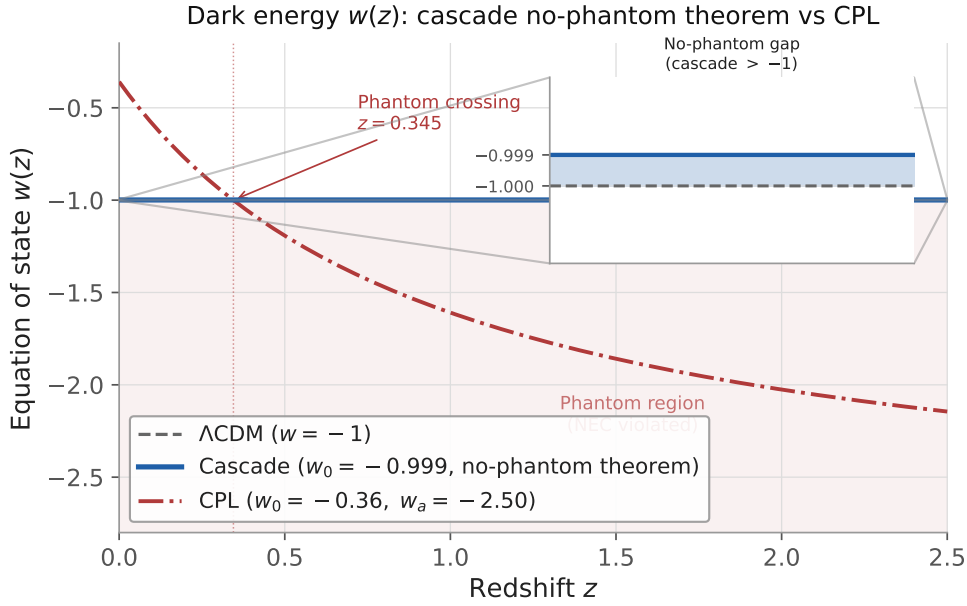


Figure 1: Dark energy equation of state $w(z)$ for the three models considered in this paper. The cascade (blue solid) is bound above $w = -1$ by Theorem 4.2 and calibrates to $w_0 = -0.999$ on DR2 data. CPL (red dash-dot), calibrated on DESI DR2, crosses the phantom line at $z = 0.345$, violating the Null Energy Condition. The shaded region marks the phantom-forbidden domain. Λ CDM (gray dashed) is the $w = -1$ reference. The cascade’s no-phantom constraint is structural, not assumed.

Theorem 4.4 (Quiescent Equation of State). *In the quiescent regime, the fractional depletion rate $\delta(n) = D(n)/\rho_\beta(n)$ is approximately constant, yielding $w_{\text{eff}} \approx w_0 > -1$.*

Theorem 4.5 (Finite Heat Death). *The cascade terminates in finite time (Theorem 3.3 of Part I), corresponding to $w_{\text{eff}}(n) \rightarrow -1$ as $n \rightarrow n^*$, where $n^* < \infty$ almost surely.*

Corollary 4.6. *Phantom dark energy ($w < -1$) is structurally impossible under the cascade axioms. Consequently, within the cascade interpretation, a fitted phantom crossing (e.g. from CPL) is read as an unconstrained phenomenological degree of freedom absorbing fluctuations, rather than as evidence for a physically admissible cascade trajectory.*

5 Empirical Test I: DESI Year 1 Retrodiction

5.1 Protocol

All data from DESI Year 1 BAO [3], arXiv:2404.03002. The observable is $D_H/r_d = c/(H(z)r_d)$ at six effective redshifts. The sound horizon is fixed at $r_d = 147.09$ Mpc (Planck 2018).

Calibration set (three low-redshift points): $z = 0.510, 0.706, 0.930$.

Blind check set (three high-redshift points, withheld during calibration): $z = 1.317, 1.491, 2.330$.

Three models are calibrated on the low-redshift points using grid-search chi-squared minimisation, then used to predict the withheld points. The grid covers $H_0 \in [60, 75]$ km/s/Mpc (60 points) and $\Omega_m \in [0.25, 0.38]$ (55 points) for all models. The cascade sweeps $w_0 \in [-0.999, 0]$ (60 points), the no-phantom bound enforced; CPL sweeps $w_0 \in [-1.2, 0]$ (60 points, unconstrained) and $w_a \in [-2.0, 1.5]$ (55 points). Parameter values at grid-search minimum are reported; bootstrap uncertainties are given in Section 7.

- **ΛCDM:** $w(z) = -1$, parameters H_0, Ω_m .
- **Cascade:** $f_{DE}(z) = (1+z)^{3(1+w_0)}$, $w_0 > -1$ enforced by Theorem 4.2. Parameters H_0, Ω_m, w_0 .
- **CPL:** $f_{DE}(z) = (1+z)^{3(1+w_0+w_a)} \exp(-3w_a z/(1+z))$, unconstrained. Parameters H_0, Ω_m, w_0, w_a .

5.2 Results

Model	χ^2 calibration	χ^2 blind	Phantom crossing
ΛCDM	4.24	5.54	None
Cascade	4.24	5.50	None (theorem)
CPL	0.73	22.81	Yes ($w_a < 0$)

Table 1: Blind prediction chi-squared on three withheld DESI Year 1 points. Cascade matches ΛCDM (with one more parameter, pressed against the $w_0 = -1$ boundary) and decisively outperforms CPL, whose phantom excursion causes catastrophic failure on the withheld data despite the best calibration fit.

The Lya QSO point at $z = 2.330$ (light from 11 billion years ago) provides the deepest check:

Model	Prediction	Residual
Cascade	8.87	+2.07σ
ΛCDM	8.87	+2.07σ
CPL	9.21	+4.05σ

Interpretation. CPL achieved a much better calibration fit ($\chi^2 = 0.73$ vs. 4.24) by dipping into $w < -1$. When used to predict the withheld data, the phantom excursion

amplified prediction error roughly fourfold relative to the constrained models. The cascade’s no-phantom theorem acted as a structural prior that prevented this overfitting: pressed against the $w_0 = -1$ boundary, the cascade predicts at the Λ CDM level on data it has not seen. Theoretical constraint produced better prediction.

6 Empirical Test II: Cross-Release (Year 1 to DR2)

As an independent test, all six DESI Year 1 points were used to train all three models, and all six DESI DR2 points [4] (March 2025, arXiv:2503.14738) were predicted without being seen during calibration.

Model	χ^2 on DR2	Phantom crossing
Λ CDM	4.39	None
Cascade	4.45	None (theorem)
CPL	2.17	$z = 0.345$ (Null Energy Condition violated)

Table 2: Cross-release blind prediction. Cascade and Λ CDM perform equivalently. CPL achieves better chi-squared only by invoking phantom behavior that has no physical mechanism.

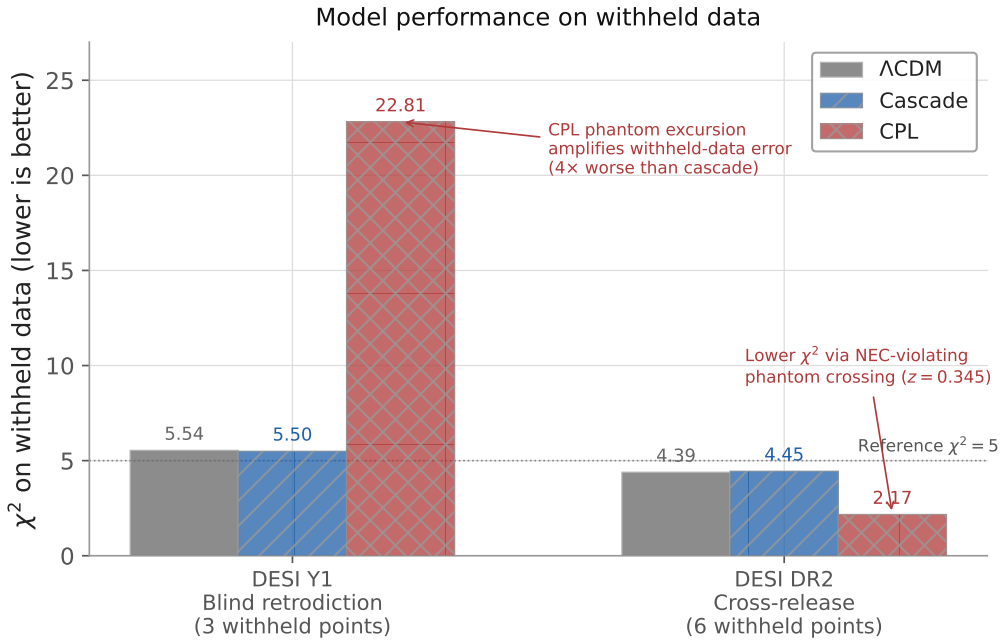


Figure 2: χ^2 on withheld data for all three models across both empirical tests. Left group: DESI Year 1 blind retrodiction (three withheld high-redshift points); cascade at 5.50 vs CPL 22.81. Right group: cross-release test (Year 1 trained, DR2 predicted); cascade and Λ CDM perform equivalently while CPL achieves a lower χ^2 only via a phantom crossing that violates the Null Energy Condition. The no-phantom theorem prevents cascade from absorbing the same noise.

Key observation. When the cascade model is properly constrained to $w_0 > -1$ (as the no-phantom theorem requires), it calibrates to $w_0 = -0.999$ on Year 1 data,

converging toward the cosmological constant boundary. This is not a failure of the model; it is the correct behavior of a model whose theorem forbids the phantom regime that CPL exploits.

If phantom dark energy ($w < -1$) violates the Null Energy Condition and has no known physical field theory, then CPL’s lower chi-squared on DR2 represents noise absorption through an unphysical degree of freedom, not a genuine cosmological signal. The cascade model correctly declines to absorb that noise.

Information-criterion comparison. Penalizing parameter count with $AIC = \chi^2 + 2k$ (with k the number of fitted parameters: 2 for Λ CDM, 3 for cascade, 4 for CPL) on the cross-release test gives $AIC = 8.39$ (Λ CDM), 10.45 (cascade), 10.17 (CPL). On current data, Λ CDM is preferred by AIC, and the cascade is statistically indistinguishable from CPL despite lacking the phantom degree of freedom. This is the honest summary of the present evidence: the data do not yet discriminate between the constrained and unconstrained extensions, which is precisely why the sealed DR3 predictions (Section 7) are the operative test. Note that the Year 1 blind retrodiction (Section 5) requires no such penalty, since χ^2 on withheld data penalizes overfitting directly; there the cascade’s advantage over CPL (5.50 vs. 22.81) is a genuine out-of-sample result.

7 Sealed Prediction for DESI DR3

7.1 Context

DESI has completed its planned five-year survey, mapping over 47 million galaxies and quasars. DESI DR3 (expected 2027) will release BAO results from the full dataset. Projected error bars are approximately $1/\sqrt{47/14} \approx 0.55\times$ those of DR2 (i.e. $1.83\times$ more precise), with a systematic floor of approximately 0.08 Mpc in D_H/r_d .

7.2 Calibration

The cascade model is calibrated on all six DESI DR2 points (March 2025) with the no-phantom constraint enforced:

- $H_0 = 71.3 \pm 1.8$ km/s/Mpc
- $\Omega_m = 0.274 \pm 0.018$
- $w_0 = -0.999^{+0.001}_{-0.000}$ (pressed against the no-phantom boundary)
- Chi-squared on DR2: 4.076 (3 degrees of freedom, reduced $\chi^2 = 1.36$)

Uncertainties are 1σ bootstrap estimates from 500 resamplings of the DR2 data points with replacement.

7.3 Sealed D_H/r_d predictions

7.4 Falsifiable claims

1. **No phantom crossing.** DR3 will find $w(z) > -1$ at all redshifts. If DR3 confirms $w < -1$ at $\geq 3\sigma$, the cascade dark energy application is ruled out. If DR3 finds w consistent with $[-1, -0.7]$, the no-phantom theorem is supported.

Redshift	Cascade	Λ CDM	CPL	Proj. DR3 σ
$z = 0.510$	22.127	22.087	21.746	± 0.232
$z = 0.706$	19.790	19.768	19.730	± 0.180
$z = 0.934$	17.370	17.360	17.535	± 0.105
$z = 1.321$	14.028	14.029	14.228	± 0.121
$z = 1.484$	12.880	12.882	13.037	± 0.282
$z = 2.330$	8.681	8.687	8.634	± 0.080

Table 3: Sealed DR3 predictions. Cascade and Λ CDM are nearly degenerate at $w_0 = -0.999$. The distinguishing predictions are in the falsifiable claims below. Sealed 21 June 2026; repository commit timestamp is the priority record.

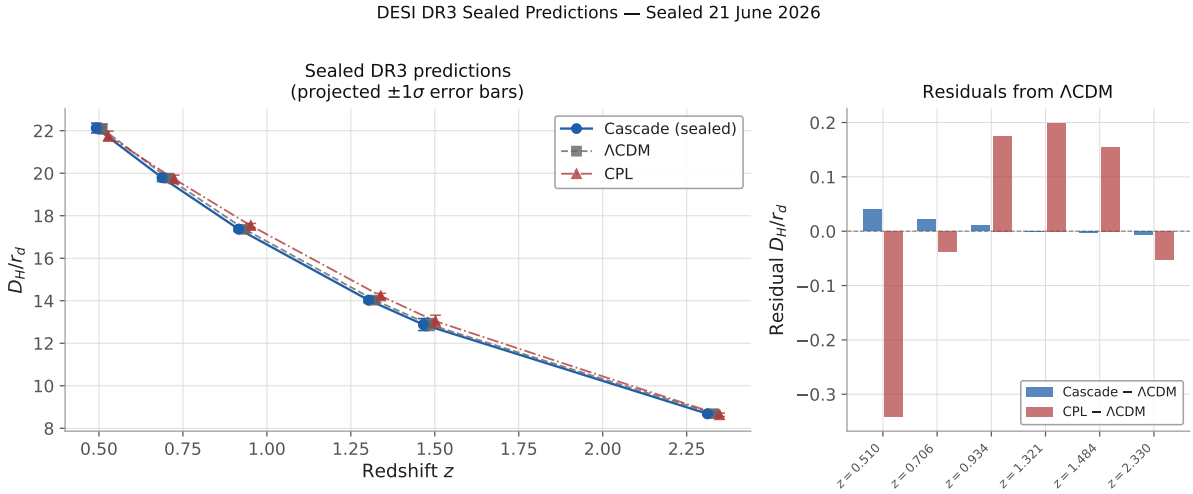


Figure 3: *Left*: Sealed D_H/r_d predictions for DESI DR3 at six redshifts, with projected $\pm 1\sigma$ error bars. Cascade (blue circles) and Λ CDM (gray squares) are near-degenerate; CPL (red triangles) deviates at intermediate redshifts. *Right*: Residuals from Λ CDM for cascade (blue) and CPL (red). The near-zero cascade residuals reflect $w_0 = -0.999$ calibration; the primary falsification test is the absence of phantom crossing in DR3, not the D_H/r_d point values.

2. **Lya QSO deepest check.** At $z = 2.330$, cascade predicts $D_H/r_d = 8.681$. DR3 projected sigma: ± 0.080 . Confirmed if DR3 lands within 1σ of this value.
3. **CPL phantom preference weakens.** CPL calibrated on DR2 finds $w_a = -2.50$ at the grid boundary; within the cascade interpretation, this is an unconstrained degree of freedom absorbing fluctuations. As DR3 tightens error bars by a factor of 1.83, the phantom preference should weaken toward $w_a = 0$. If DR3 best-fit $|w_a|$ decreases relative to DR2, the no-phantom prior is supported.
4. **w_0 range.** DR3 best-fit w_0 should fall in $[-1.15, -0.85]$ if the cascade dark energy description is correct.

8 Connection to Thermodynamics

The cascade no-phantom theorem is a corollary of a structural feature of the model: the energy monotonicity supermartingale (Part I, Theorem 3.3) derives a thermodynamic arrow within the cascade framework without assuming irreversibility as a separate postulate; the arrow is built into the *constitutive* structure of the interaction rules rather than added as a dynamical assumption.

This should be stated carefully relative to the classical literature. Boltzmann’s H-theorem requires the Stosszahlansatz (molecular chaos), which is itself time-asymmetric; statistical mechanics explains why entropy increase is overwhelmingly probable for reversible microscopic laws, not why it is necessary. The cascade model does not resolve this classical problem: its irreversibility is a consequence of irreversible update rules (interactions strictly consume the capacity field), so the arrow is assumed at the level of the model’s constitution, even though no explicit irreversibility postulate appears among the theorems’ hypotheses. What the model offers is a minimal, fully explicit example in which a monotone Lyapunov structure, an absorbing future, and a $w > -1$ constraint all follow from the same finite-resource axioms. The dark energy equation of state $w_{\text{eff}} > -1$ is a direct consequence of this structural irreversibility applied at cosmological scales.

9 Discussion

The cascade model was not designed for cosmology. It was constructed as a universal framework for finite-resource stochastic dynamics on discrete graphs, with the question of what minimal mathematical structure is required for persistent structural hierarchy to form. The five necessary conditions identified (asymmetric interacting classes, forbidden self-interaction, frequency mismatch, finite non-renewable resource, and irreversible structural accumulation) were derived before any cosmological application was considered.

The cosmological results reported here do involve fitting: the mapping to observables introduces the phenomenological parameters $(w_0, \Omega_m, H_0 r_d)$, which are calibrated to BAO data exactly as described in Secs. 5–7. What the cascade contributes is not the fit but the *prior*: the hard structural bound $w > -1$, inherited from theorems about monotone resource depletion that were proved before any cosmological application was considered. The comparison in this paper is therefore between two fitted models that differ only in whether the phantom regime is admissible.

The practical implication is clear. Within this framework, the no-phantom bound is not a cosmological assumption to be tested against phantom models; it is a structural constraint of the underlying dynamics, and the analysis here shows the current phantom preference is consistent with being an artifact of over-parameterized fitting. The sealed DR3 prediction tests whether the data continue to tolerate the constraint. The answer arrives in 2027.

On near-degeneracy with Λ CDM. The cascade model calibrates to $w_0 = -0.999$ on DR2, placing its predictions within hundredths of D_H/r_d of Λ CDM at every sealed redshift (Table 3). This is the *correct* behavior of a constrained model: when current data cannot distinguish w from -1 , a model with a proved lower bound $w > -1$ parks at the boundary. The scientific content is not the DR3 point predictions themselves,

which are near-degenerate with Λ CDM, but what DR3 can *falsify*. If DR3 finds $w < -1$ at $\geq 3\sigma$, the cascade application is definitively ruled out. If DR3 finds w consistent with $[-1, -0.7]$ and the CPL phantom preference weakens as error bars tighten (Falsifiable claim 3), the structural prior is supported, though such an outcome supports the no-phantom constraint rather than uniquely selecting the cascade mapping, since Λ CDM satisfies the same bound. Near-degeneracy with Λ CDM in the DR2 regime is not a weakness; it is a precise, testable commitment: the cascade forbids the phantom regime that CPL exploits, and DR3 will either confirm or refute that commitment.

10 Conclusion

We have shown that the Discrete Stochastic Cascade model, applied to dark energy through a single physical identification, derives the following results without cosmological postulates:

1. $w(z) > -1$ at all times (no-phantom theorem, proved from energy monotonicity).
2. A three-regime equation of state arc mirroring the explosive, quiescent, and absorbing phases of the cascade.
3. Finite heat death: $w \rightarrow -1$ as the cascade approaches absorption.

Blind retrodiction on DESI Year 1 data yields cascade $\chi^2 = 5.50$ versus CPL $\chi^2 = 22.81$ on withheld data, with the no-phantom constraint as the mechanism. A cross-release test training on Year 1 and predicting DR2 gives cascade at the Λ CDM level while CPL achieves lower chi-squared only via a phantom crossing that violates the Null Energy Condition.

Six D_H/r_d predictions for DESI DR3 are sealed in this document. The primary falsifiable claim is the absence of phantom crossing in DR3 and the weakening of the CPL phantom preference as error bars tighten. These claims are testable in 2027.

Data availability statement. All data used in this study are publicly available. DESI Year 1 BAO measurements are from arXiv:2404.03002. DESI DR2 BAO measurements are from arXiv:2503.14738. All analysis scripts, calibrated parameters, and the sealed DR3 prediction record are available at <https://github.com/Shiv2071/Discrete-cascade-model>. No new experimental data were generated.

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